

Baseball Practice Session 1

Welcome to the first practice session of the semester! These sessions are meant to practice and reinforce the material taught in lecture. We will hold these sessions once a week, and each session will be 1 hour. Note: practice sessions will NOT be turned in.

In today's practice session, you'll practice:

1. Writing and evaluating some basic *expressions* in Python
2. Using pandas to manipulate baseball data through some guided exercises

Loading the Lahman Baseball Database

```
import YData_baseball

YData_baseball.download_data("Lahman_2024u/Batting.csv")
YData_baseball.download_data("Lahman_2024u/People.csv")
```

The file `Batting.csv` already exists.

If you would like to download a new copy of the file, please rename the existing copy of the file.

The file `People.csv` already exists.

If you would like to download a new copy of the file, please rename the existing copy of the file.

Loading the pandas and numpy Packages

```
import pandas as pd
import numpy as np
```

1. Quick Review of Python Data Types and Structures

1.1 Basic data types

Some of the main types of data in Python are:

- **int**: A whole number
- **float**: A decimal number
- **bool**: A logical value that can be True or False
- **strings**: A piece of text

We can also check what type of value is in a variable using the `type()` function.

1.2 Basic data structures

Some of the main data structures in Python are:

- **List**: `my_list = [1, 2, 3]`
- **Tuple**: `my_tuple = (1, 2, 3)`
- **Dictionary**:

```
my_dictionary = {  
    "key1": "value1",  
    "key2": "value2",  
    "key3": "value3"  
}
```

1.3 Pandas DataFrames

The most applicable data structure for data science is the DataFrame, which is basically a spreadsheet in Excel. The Pandas package allows us to use DataFrames. Here is the players DataFrame. We can look at the first five rows by adding `.head(5)` to the end.

```
batting = pd.read_csv("Batting.csv")  
batting.head(5)
```

	playerID	yearID	stint	teamID	lgID	G	G_batting	AB	R	H	...	SB	CS	BB	SO	IB
0	aardsda01	2004	1	SFN	NL	11	NaN	0	0	0	...	0.0	0.0	0	0.0	0.0
1	aardsda01	2006	1	CHN	NL	45	NaN	2	0	0	...	0.0	0.0	0	0.0	0.0
2	aardsda01	2007	1	CHA	AL	25	NaN	0	0	0	...	0.0	0.0	0	0.0	0.0

	playerID	yearID	stint	teamID	lgID	G	G_batting	AB	R	H	...	SB	CS	BB	SO	IB
3	aardsda01	2008	1	BOS	AL	47	NaN	1	0	0	...	0.0	0.0	0	1.0	0.0
4	aardsda01	2009	1	SEA	AL	73	NaN	0	0	0	...	0.0	0.0	0	0.0	0.0

1.4 Numpy Arrays

Numpy arrays are similar to lists, but all elements in a numpy array must be of the same type (e.g., all integers, all strings).

2. Data Structure Practice

2.1 Creating a dictionary

Create a dictionary that maps each of the following abbreviations to their full name: 'G', 'G_batting', 'AB', 'R', 'H', '2B', '3B', 'HR', 'RBI', 'SB', 'CS', 'BB', 'SO', 'IBB', 'HBP', 'SH', 'SF', and 'GIDP'.

```
# Your Code Here
```

2.2 Creating a Numpy Array

Convert the following lists to arrays. Then calculate the walks per plate appearance. Round your answer to two decimal places.

```
import numpy as np

# Lists showing Mike Trout's seasons, walks, and plate appearances
seasons = [2014, 2015, 2016, 2017, 2018]
walks = [83, 92, 116, 94, 122]
plate_appearances = [705, 682, 681, 507, 608]

# Your Code Here
```

2.3 Selecting Columns from a DataFrame

Consider the players DataFrame. Select the columns `playerID`, `nameGiven`, `birthYear`, and `birthState`.

```
players = pd.read_csv("People.csv")  
  
# Your Code Here
```

2.4 Subsetting Rows in a DataFrame

Looking again at the players DataFrame, write code to select the players that were born in Denver.

```
# Your Code Here
```

2.5 Using Methods on Columns in a DataFrame

The pandas Series and DataFrames also have methods that can operate on the values in a column. For example, we can sum the values in a column using the `my_df.sum()` method. We can also use numpy functions on columns of a DataFrame if we extract the data in the columns as pandas Series. For we can also sum the values in a column of a pandas DataFrame using `np.sum(my_df["col_name"])`.

Looking back at the batting data, what is the total number of stolen bases from 1871 through the 2024 season? Please calculate this value using both the pandas `.sum()` method and the numpy `np.sum()` function

```
import numpy as np  
  
# Your Code Here
```

2.6 Boolean indexing and the `.query()` methods

As we discussed extensively in YData, you can use Boolean indexing and/or the `.query()` method to extract a DataFrame that only contains rows that meet particular conditions. For example, if we have a DataFrame called `my_df` that has a column called `vals` that has quantitative data, you could extract a DataFrame that only has rows where `vals` are greater than 10 using either:

- `my_df[my_df["vals"] > 10]`
- `my_df.query("vals > 10")`

2.7 Aggregate statistics by group

We can use the pandas `groupby()` and `.agg()` method to calculate aggregate statistics separately for different groups. For example, we can have a DataFrame called `my_df` that had a categorical column called “`group_col`” and a quantitative column called “`quant_col`”, we could calculate the sum of the values in the “`quant_col`” separately for each unique value in the “`group_col`” using `my_df.groupby("group_col").agg(sum_vals = ("quant_col", "sum"))`

Using the batting data, please calculate the average (“mean”) number of strike outs for each season from 1871 to 2024. Also, sort the values such that the season that has the most strike outs is at the top of this aggregated DataFrame.

```
# Your Code Here
```

3. Player Case Study: Mike Trout

Mike Trout is an American MLB player with an elite regular-season resume. He won the American League MVP award three times (2014, 2016, 2019), and is an 11-time All-Star. Our analysis will focus on these three seasons, and see how they compare to the league average.

3.1 Pulling Season Data

Subset the batting data to look at the 2014, 2016, and 2019 season. Save this to a DataFrame called `batting_14_16_19`.

```
# Your Code Here
```

3.2 Pulling Trout’s MVP Data

Using the `batting_14_16_19` data, filter the data to look at Trout. Note his `playerID` is `troutmi01`.

```
# Your Code Here
```

3.3 Calculating League-Wide Summary Statistics

Using the Numpy `.mean()` method and the `batting_14_16_19` data, calculate the average runs, hits, home runs, stolen bases, and RBIs for the league.

```
# Your Code Here
```

3.4 Calculating Trout's Summary Statistics

Repeat the above analysis using the Trout data.

```
# Your Code Here
```

3.5 Comparing Trout to the League

Compare Trout's performance to that of the league. What are pros to this analysis? What are some cons?

```
%%capture  
  
# You can run this code to covert this Jupyter notebook into a pdf  
!quarto render practice_1.ipynb --cache-refresh --to pdf
```